DEVOPS - 03-03-2026

```
C:\Users\HP>notepad dockerfile
```

Itha dockerfile notepad la dockerfile.txt
(build pannum pothu error varum so rename pannitom using ren command below iruku)

```
Dockerfile
File    Edit    View    H1

FROM ubuntu:latest
RUN apt update
RUN apt install apache2 -y
COPY index.html /var/www/html
RUN apt install systemctl -y
RUN systemctl start apache2
RUN systemctl enable apache2
ENTRYPOINT apachectl -D FOREGROUND
```

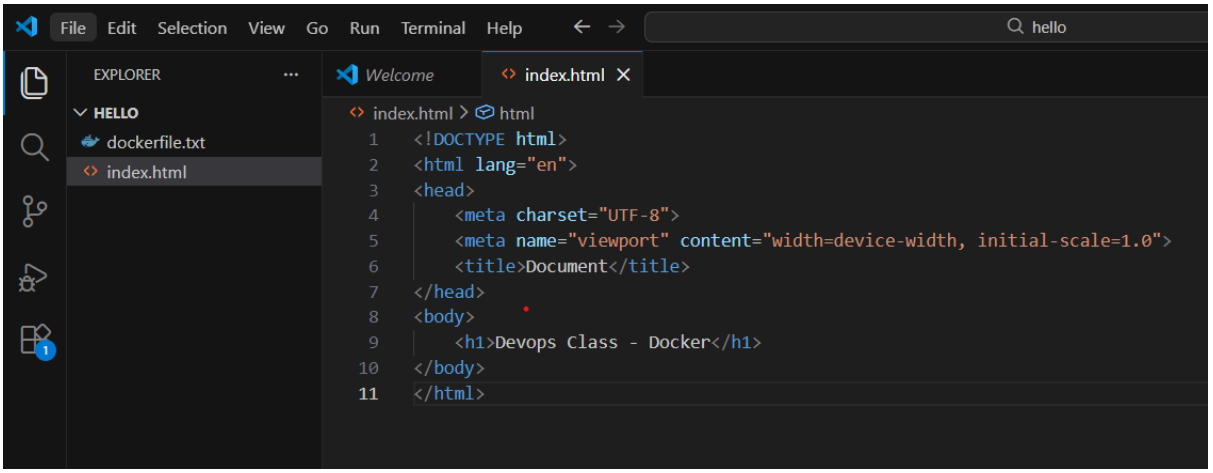
Write this command in Dockerfile

```
C:\Users\HP>mkdir hello
```

Oru directory create pandrom for accessing so ithu la intha Dockerfile and creating a index.html file also

```
C:\Users\HP>notepad index.html
```

Ithu index.html create panna else create manually



Index.html file

```
C:\Users\HP>cd hello
```

moving to created directory

```
C:\Users\HP\hello>ren Dockerfile.txt Dockerfile
```

(build error vandatuku aprm file name change pannitom)

```
C:\Users\HP\hello>docker build .
[+] Building 214.3s (12/12) FINISHED                    docker:desktop-linux
 => [internal] load build definition from Dockerfile                    0.0s
 => => transferring dockerfile: 257B                                    0.0s
 => [internal] load metadata for docker.io/library/ubuntu:latest        0.3s
 => [internal] load .dockerignore                                       0.1s
 => => transferring context: 2B                                         0.0s
 => [internal] load build context                                       0.1s
 => => transferring context: 284B                                       0.0s
 => [1/7] FROM docker.io/library/ubuntu:latest@sha256:d1e2e92c075e5ca   0.2s
 => => resolve docker.io/library/ubuntu:latest@sha256:d1e2e92c075e5ca   0.1s
 => [2/7] RUN apt update                                               34.4s
 => [3/7] RUN apt install apache2 -y                                  141.0s
 => [4/7] COPY index.html /var/www/html                                 0.1s
 => [5/7] RUN apt install systemctl -y                                 22.1s
 => [6/7] RUN systemctl start apache2                                   4.9s
 => [7/7] RUN systemctl enable apache2                                  1.0s
 => exporting to image                                                  9.7s
 => => exporting layers                                                 6.7s
 => => exporting manifest sha256:2bd3c48bca3613fa0bac8f47e5e45bfb2822   0.0s
 => => exporting config sha256:302ff1008087cabb2a77c63eafd865bef784a1   0.0s
 => => exporting attestation manifest sha256:1ff5a847e8196ef849d327c7   0.1s
 => => exporting manifest list sha256:2897b4a74aeb900a5f65bbe2aad2450   0.1s
 => => naming to moby-dangling@sha256:2897b4a74aeb900a5f65bbe2aad2450   0.0s
 => => unpacking to moby-dangling@sha256:2897b4a74aeb900a5f65bbe2aad2   2.7s

 1 warning found (use docker --debug to expand):
 - JSONArgsRecommended: JSON arguments recommended for ENTRYPOINT to prevent
 unintended behavior related to OS signals (line 8)

C:\Users\HP\hello>
```

Used to build the file

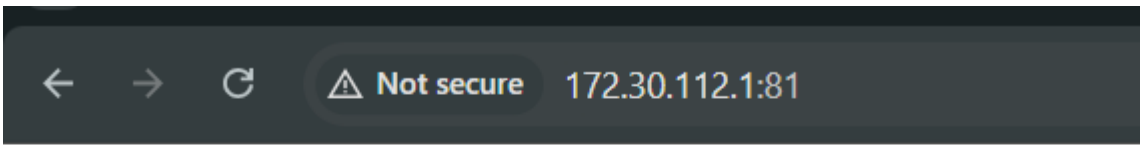
```
C:\Users\HP>docker build -t my-img -f c:/Users/HP/hello/Dockerfile .
[+] Building 1.9s (8/11)                                  docker:desktop-linux
 => [internal] load build definition from Dockerfile                      0.0s
 => => transferring dockerfile: 257B                                      0.0s
 => [internal] load metadata for docker.io/library/ubuntu:latest          1.3s
 => [internal] load .dockerignore                                         0.1s
 => => transferring context: 2B                                           0.0s
 => [internal] load build context                                         0.1s
 => => transferring context: 2B                                           0.0s
 => [1/7] FROM docker.io/library/ubuntu:latest@sha256:d1e2e92c075e5ca  0.1s
 => => resolve docker.io/library/ubuntu:latest@sha256:d1e2e92c075e5ca  0.1s
 => CACHED [2/7] RUN apt update                                           0.0s
 => CACHED [3/7] RUN apt install apache2 -y                               0.0s
 => ERROR [4/7] COPY index.html /var/www/html                             0.0s
------
 > [4/7] COPY index.html /var/www/html:
------

 1 warning found (use docker --debug to expand):
 - JSONArgsRecommended: JSON arguments recommended for ENTRYPOINT to prevent
 unintended behavior related to OS signals (line 8)
Dockerfile:4
--------------------
   2 |     RUN apt update
   3 |     RUN apt install apache2 -y
   4 | >>> COPY index.html /var/www/html
   5 |     RUN apt install systemctl -y
   6 |     RUN systemctl start apache2
--------------------
ERROR: failed to build: failed to solve: failed to compute cache key: failed
 to calculate checksum of ref ngf2mqzj9hlkx1lk5paxl2cda::lyien67zgqzvo932ypc
6426h2: "/index.html": not found

C:\Users\HP>
```

```
C:\Users\HP\hello>docker build -t my-img .
[+] Building 0.7s (12/12) FINISHED                    docker:desktop-linux
 => [internal] load build definition from Dockerfile                    0.0s
 => => transferring dockerfile: 257B                                    0.0s
 => [internal] load metadata for docker.io/library/ubuntu:latest        0.1s
 => [internal] load .dockerignore                                       0.0s
 => => transferring context: 2B                                         0.0s
 => [1/7] FROM docker.io/library/ubuntu:latest@sha256:d1e2e92c075e5ca   0.0s
 => => resolve docker.io/library/ubuntu:latest@sha256:d1e2e92c075e5ca   0.0s
 => [internal] load build context                                       0.0s
 => => transferring context: 32B                                        0.0s
 => CACHED [2/7] RUN apt update                                         0.0s
 => CACHED [3/7] RUN apt install apache2 -y                             0.0s
 => CACHED [4/7] COPY index.html /var/www/html                          0.0s
 => CACHED [5/7] RUN apt install systemctl -y                           0.0s
 => CACHED [6/7] RUN systemctl start apache2                            0.0s
 => CACHED [7/7] RUN systemctl enable apache2                           0.0s
 => exporting to image                                                  0.2s
 => => exporting layers                                                 0.0s
 => => exporting manifest sha256:2bd3c48bca3613fa0bac8f47e5e45bfb2822   0.0s
 => => exporting config sha256:302ff1008087cabb2a77c63eafd865bef784a1   0.0s
 => => exporting attestation manifest sha256:a144a215e21de1f3f701fcc5   0.1s
 => => exporting manifest list sha256:ff9cbfaaebd27824772a2c70a69c293   0.0s
 => => naming to docker.io/library/my-img:latest                        0.0s
 => => unpacking to docker.io/library/my-img:latest                     0.0s

 1 warning found (use docker --debug to expand):
 - JSONArgsRecommended: JSON arguments recommended for ENTRYPOINT to prevent
 unintended behavior related to OS signals (line 8)

C:\Users\HP\hello>
```

```
C:\Users\HP\hello>docker images
                                                  i Info →   U  In Use
IMAGE                        ID             DISK USAGE   CONTENT SIZE   EXTRA
admin/customwebserver:1.0    2fa91ad76a61        279MB         76.6MB
httpd:latest                 96b1e8f69ee3        177MB         47.6MB    U
my-img:latest                ff9cbfaaebd2        424MB          118MB
sietbk:1.0                   2fa91ad76a61        279MB         76.6MB
ubuntu:latest                d1e2e92c075e        119MB         31.7MB    U
```

```
C:\Users\HP\hello>docker run -d -it --name my-website -p 81:80 my-img:latest

9370b30c1c89902c142e0cae680c0fd13157f1cd689dde30eafdb8f1add7338d
```

```
C:\Users\HP\hello>ipconfig
```



Devops Class - Docker

DOCKER NETWORK

List of network to view use this command. Default 3 network will have.

```
C:\Users\HP\hello>docker network ls
NETWORK ID     NAME      DRIVER    SCOPE
8e31ef174e8c   bridge    bridge    local
b7a2fd77fa43   host      host      local
3d303eb80444   none      null      local
```

Creating a network

```
C:\Users\HP\hello>docker network create new-network
9628994c28e46590fd421d0da97d259951194357740fb03fbbf0f4841094d61e
```

```
C:\Users\HP\hello>docker network ls
NETWORK ID     NAME          DRIVER    SCOPE
8e31ef174e8c   bridge        bridge    local
b7a2fd77fa43   host          host      local
9628994c28e4   new-network   bridge    local
3d303eb80444   none          null      local
```

Now the created network is in the list of networks